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### Self-service platform integration

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#### Abstract

Systems include reception of a first set of data from a first posting platform, determination of a plurality of field definitions based on the first set of data, each field definition including a field and attributes associated with the field, transmission of the plurality of field definitions to the first posting platform, reception of one or more modifications to the plurality of field definitions from the first posting platform, generation of a second plurality of field definitions based on the one or more modifications to the plurality of field definitions, and storage of the second plurality of field definitions.

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## Background/Summary

### BACKGROUND

(1) Modern software applications enable efficient execution of tasks. Such tasks may include generation, storage and analysis of data records, and communication with other applications or services based on such data records. For example, a user may operate an application to create a record, to select one or more websites, and to transmit the record to the one or more selected websites. Each of the selected websites may be configured to receive data records in a particular format, and the application therefore formats a data record in accordance with a respective particular format prior to transmission to a website.

(2) Establishing and maintaining the technical details (data structure, protocol, etc.) which govern communication between an application and various third-party platforms such as websites may include multiple stakeholders and manual actions, and thereby consume an unacceptable amount of time and resources. In one example, a recruiting application may include software modules which enable a user to create a job requisition, to select multiple third-party job posting platforms (i.e., job boards), and to transmit the job requisition to the selected job boards. In order to enable the recruiting application to transmit a job requisition to a particular job board, the recruiting application must be configured to transmit the job requisition in conformance with technical specifications which govern communication with the job board.

(3) Configuration typically begins with a request from a user to integrate a new job board with the recruiting application. A provider of the recruiting application prioritizes the request within a queue and, once the request rises to the top of the queue, the provider discusses technical specifications with job board personnel. Agreement on the technical specifications is reached, and the development activities which are required to configure the recruiting application based on the

technical specifications are queued based on priority. Once the development activities are performed, the configuration is tested by the provider of the recruiting application and validated by the job board. The thusly-configured application is placed into production if testing and validation are successful.

(4) The foregoing process must occur independently with respect to each job board to which a recruiting application intends to transmit job requisitions. This process may therefore require an unacceptable amount of time and resources. These issues are exacerbated in the case of certain types of third-party platforms, such as job boards, which are rapidly increasing in number and complexity. Systems are therefore desired to facilitate integration between an application and a third-party platform.

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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

(1) FIG. 1 is a block diagram of an architecture to integrate heterogeneous posting platforms with a posting application according to some embodiments.

(2) FIG. 2 comprises a flow diagram of a process to integrate a posting platform with a posting application according to some embodiments.

(3) FIG. 3 is a view of a user interface to provide a markup language sample according to some embodiments.

(4) FIG. 4 is a view of a user interface to define markup language fields and associated attributes according to some embodiments.

(5) FIG. 5 is a view of a user interface to review and confirm a markup language template according to some embodiments.

(6) FIG. 6 is a view of a user interface to specify a data transfer protocol according to some embodiments.

(7) FIG. 7 is a block diagram of an architecture including a recruiting application and a plurality of job boards integrated therewith according to some embodiments.

(8) FIG. 8 comprises a flow diagram of a process executable by a recruiting application to post a job requisition to a job board integrated therewith according to some embodiments.

(9) FIG. 9 is a view of a user interface to specify a job requisition for posting to a job board according to some embodiments.

(10) FIG. 10 is a view of a user interface to specify one or more job boards to which a job requisition is to be posted according to some embodiments.

(11) FIG. 11 is a view of a user interface for completing fields associated with job boards according to some embodiments.

(12) FIG. 12 is a block diagram of a hardware system for providing a recruiting application according to some embodiments.

### **DETAILED DESCRIPTION**

(13) The following description is provided to enable any person in the art to make and use the described embodiments and sets forth the best mode contemplated for carrying out some embodiments. Various modifications, however, will be readily-apparent to those in the art.

(14) Some embodiments relate to systems which facilitate integration between an application and heterogeneous external platforms (e.g., websites, portals, data aggregators). According to some implementations, an integration service is provided by which a third party may initiate and substantially configure an application for communication with an external platform while reducing a need for corresponding manual actions by a provider/developer of the application. Embodiments may therefore conserve development and management resources while speeding the establishment of integrations between an application and external platforms. For users of the application,

embodiments may increase transparency surrounding such an integration because the users may simply request a third-party platform to initiate integration with the application, without having to unduly rely on the provider/developer of the application to do so.

(15) FIG. 1 is a block diagram of an architecture of system **100** according to some embodiments. The illustrated elements of system **100** may be implemented using any suitable combination of computing hardware and/or software that is or becomes known. In some embodiments, two or more elements of system **100** are implemented by a single computing device. Two or more elements of system **100** may be co-located. One or more elements of system **100** may be implemented as a cloud service (e.g., Software-as-a-Service, Platform-as-a-Service). A cloud-based implementation of any elements of system **100** may apportion computing resources elastically according to demand, need, price, and/or any other metric.

(16) Application platform **110** may comprise an on-premise, cloud-based, or hybrid computing platform providing an execution platform and services to software applications. Application platform **110** may comprise one or more virtual machines executing program code of an application server. The software applications may comprise program code executable by one or more processing units (e.g., Central Processing Units (CPUs), processor cores, processor threads) of application platform **110** to provide various functions.

(17) Posting platforms **120**, **122** and **124** may each comprise an on-premise, cloud-based or hybrid computing system which may accept data from one or more applications executing on application platform **110**. Such data will be referred to herein as “posts” and may comprise any data that is or becomes known. Posting application **112** may provide functionality for transmitting such data (i.e., posting posts) to selected ones of posting platforms **120**, **122** and **124**. According to some embodiments, posts received by one of posting platforms **120**, **122** and **124** from posting application **112** may be viewed by users of the posting platform. The users, the provider of the posting platform, and the provider of posting application **112** may be unrelated entities. Specific embodiments thereof will be described below.

(18) Two or more of posting platforms **120**, **122** and **124** may have different technical requirements associated with posts received from posting application **112** (or from any other posting system). That is, input data to be received and posted by posting platform **120**, for example, must be in a different format and/or transmitted via a different protocol than input data to be received and posted by posting platform **122**. Posting platform **124** may require input data of yet a third format and/or protocol.

(19) Application platform **110** also includes program code of integration service **113**. Although illustrated separately in FIG. 1, integration service **113** may comprise a component, module, etc. of posting application **112** according to some embodiments. Integration service **113** allows a third party (i.e., different from a provider of posting application **112**) to specify technical and other details of a posting platform. The technical details will be referred to as a profile and may include a data structure, attributes of various fields of the data structure, a transmission protocol, a name, a logo, contact details, etc. In some embodiments, integration service **113** provides a portal which may be accessed over the Web.

(20) FIG. 1 shows direct communication between integration service **113** and posting platforms **120**, **122** and **124**. In some embodiments, a developer of a posting platform connects (i.e., logs on) to the posting platform in an administrator role and accesses integration service **113** via the posting platform. It should be noted that integration service **113** may be accessed directly by a separate computing device/system. For example, a developer of posting platform **122** may execute a Web browser on a laptop/desktop computer to access a Web portal provided by integration service **113** and provide profile information thereto, independent of posting platform **122**.

(21) Platform profiles **114** include profiles provided by posting platforms via integration service **113**. Platform profiles **114** may be stored in any suitable format within data store **115**. Data store **115** may comprise any suitable storage system such as a database system, which may be partially or

fully remote from compute elements of platform **110**, and may be distributed as is known in the art. (22) As further shown in FIG. **1**, user **130** may access post management application **116** executing on application platform **110**. Post management application **116** may allow user **130** to compose a post for posting to one or more posting platforms. As will be described below, such a post may comprise a job requisition, but embodiments are not limited to any particular type of data. Post management application **116** may store posts in posts **117** of data store **115**, and read, update and delete posts therefrom.

(23) User **130** may further operate post management application **116** to select, via posting application **112**, one or more posting platforms to which a post should be transmitted. In some embodiments, user **130** may access posting application **112** directly to select a post of posts **117** and one or more posting platforms to which to transmit the selected post. In either case, posting application **112** identifies a platform profile of platform profiles **114** which corresponds to each selected posting platform and transmits the post to each selected posting platform based on the corresponding profile.

(24) Transmission of a post to a posting platform may comprise formatting the post according to the technical formatting details of a corresponding one of platform profiles **115** and transmitting the post according to the transmission details of the corresponding platform profile. In a case that two selected posting platforms are associated with different platform profiles, transmission of a post to each selected posting platform may require posting application **112** to format the post according to two different sets of technical formatting details and to transmit the post via two different transmission protocols.

(25) Post management application **116** and posting application **112** may comprise elements of a suite of applications provided by platform **110**. Each application in the suite may communicate with other applications of the suite via respective Application Programming Interfaces (APIs). In this example, user **130** may access a landing Web page of the suite of applications, provide suitable credentials, and then be presented with an interface allowing selection of post management application **116**, posting application **112**, and any other applications of the suite.

(26) FIG. **2** comprises a flow diagram of process **200** to integrate a posting platform with a posting application according to some embodiments. Process **200** will be described with respect to the elements of system **100**, but embodiments are not limited thereto.

(27) Process **200** and all other processes mentioned herein may be embodied in processor-executable program code read from one or more of non-transitory computer-readable media, such as, for example, a hard disk drive, a volatile or non-volatile random access memory, a DVD-ROM, a Flash drive, and a magnetic tape, and then stored in a compressed, uncompiled and/or encrypted format. A processor may include any number of microprocessors, microprocessor cores, processing threads, or the like. In some embodiments, hard-wired circuitry may be used in place of, or in combination with, program code for implementation of processes according to some embodiments. Embodiments are therefore not limited to any specific combination of hardware and software.

(28) Process **200** may be initiated by a customer of a posting application. For example, the customer (e.g., user **130** in FIG. **1**) may wish to operate a posting application (e.g., posting application **112**) to post data to a posting platform (e.g., **120**, **122** or **124**) with which the posting application is currently not able to communicate. The customer may therefore contact a provider of the posting platform and request the posting platform to initiate an integration with the posting application. In another example, a provider of a posting platform may independently initiate an integration with the posting platform. For example, a respective administrator of each posting platform **120**, **122** and **124** may interact with application platform **110** such that after completion of process **200** by each administrator, each platform **120**, **122** and **124** will be better able to communicate more efficiently with customers (not shown in FIG. **1**) seeking services from the various platforms. Embodiments are not limited to these examples.

(29) Process **200** begins at S205, at which communication with an integration service (e.g.,

integration service **113**) is established. In this regard, the integration service may include a set of Web pages associated with a publicly-accessible Uniform Resource Locator (URL). The Web pages may be used to receive and provide information as described below. **S205** may comprise operation of a computing device operated by a provider or administrator of a posting platform to execute a Web browser and direct the Web browser to the URL of the integration service. Process **200** may utilize any other suitable metaphor for exchanging information with an integration service.

(30) At **S210**, one or more markup language files are transmitted from the posting platform to the integration service. According to some embodiments, the markup language files are eXtended Markup Language (XML) files which conform to a preferred and/or required input data format of the posting platform. The one or more files may be provided in any known manner, including but not limited to File Transfer Protocol (FTP) and upload to a secure cloud-based storage location.

(31) Field definitions are received by the posting platform from the integration service at **S215**. The field definitions are determined by the integration service based on the provided one or more markup language files. Each field definition specifies a field and one or more attributes of the field. The field definitions facilitate configuration of an XML template which may be used as described above.

(32) According to some embodiments, the posting platform provider may modify the received field definitions at **S220**. The modifications may modify, add or remove fields, or modify, add or remove attributes for a given field. Modification of the field definitions and attributes may also include specification of a pick-list of allowed values for one or more of the fields.

(33) The posting platform defines a transfer protocol for receiving posting data from the integration service at **S225**. Next, at **S230**, the posting platform requests the integration service to store of the field and attribute definitions and transfer protocol.

(34) FIG. 3 illustrates user interface **310** for providing one or more markup language files to an integration service at **S210** according to some embodiments. User interface **310** may comprise a Web page provided by the integration service to a Web browser operated by a provider of a posting platform as described above. For example, an administrator of a posting platform may access an administrator home page of the posting platform, which in turn includes a hyperlink associated with the integration service and selectable to initiate integration of a posting platform.

(35) Selection of such a hyperlink (unshown) may result in display of user interface **310** by the Web browser. User interface **310** includes several tabs **320** comprising a “smart template wizard”. The smart template wizard is intended to generate a markup language template to be associated with the posting platform (e.g., within a platform profile) and to be used by the posting application to format data and transmit the formatted data to the posting platform.

(36) User interface **310** of FIG. 3 displays the contents of the SampleXML tab **320**. SampleXML tab **320** includes input area **330** for inputting a sample markup language file which conforms to an input format suitable to the posting platform. In some embodiments, a posting platform provider simply pastes an example XML feed into input area **330**. SampleXML tab **320** may be configured to accept multiple markup language files in some embodiments. The posting platform administrator may select Generate SmartXML control **340** to submit the markup language file(s) to the integration service at **S210**.

(37) FIG. 4 shows UI **311** displaying SmartXML tab **320**. SmartXML tab **320** includes form **350** presenting fields definitions at **S215** according to some embodiments. Form **350** specifies fields **351** and, for each field **351**, XML tag name **352** and other attributes **353**. As shown in FIG. 4, Display attribute **353** indicates whether the field should be displayed to a user of the posting application when posting to posting platform, the CData attribute **353** indicates whether the field includes XML character data, and the i18n attribute **353** indicates whether data of the field should be subjected to internationalization and localization. Embodiments are not limited to attributes **353**. For example, an attribute may indicate whether a field is required (i.e., a job posting based on these field definitions must include a value for the field) or optional (i.e., the job posting may include a

value for the field, but a value if not required).

(38) According to some embodiments, the integration service includes a parser to parse the provided one or more markup language files, determine field definitions therefrom, and generate form **350** based thereon. The parser parses the one or more markup language files to identify fields within the files and to identify attributes of the fields based on the markup language files. Form **350** is then generated with checkboxes associated with various attributes **353** pre-checked based on the determined fields and attributes.

(39) According to some embodiments, the posting platform provider may modify form **350** at **S220** to modify, add or remove fields, and to modify, add or remove attributes for a given field (e.g., via the checkboxes associated with attributes **353**). Once the fields and attributes are satisfactorily defined, the provider may select Generate Template control **360**.

(40) After receiving confirmation from the administrator via selection of Generate Template control **360**, the integration service returns UI **312** to the administrator as shown in FIG. 5. Template tab **320** primarily displays template **370**, which may have been generated based on the field and attribute definitions of form **350**. Template **370** specifies the format of an XML feed for input to the posting platform. Template tab **320** also includes controls **380** for specifying features of and editing template **370**.

(41) Sending method tab **320** allows definition of a transfer protocol at **S225** in some embodiments. Sending method tab **320** of FIG. 6 provides five protocol options but embodiments are not limited thereto. Sending method tab **320** also allows specification of a token to identify the posting application to the posting platform and an optional client parameter which allows the posting platform to identify posts received from a posting application on behalf of different customers of the posting application.

(42) Save control **390** initiates a request to store the field and attribute definitions and transfer protocol at **S230**. With respect to FIG. 1, selection of Save control **390** may cause integration service **113** to store a corresponding platform profile **114** in data store **115**. The platform profile **114** may include the field and attribute definitions and transfer protocol information, and may also include, for example, a platform name, a platform logo, a list of supported countries and languages, a description (e.g., in several languages), a sales contact, and a technical contact. Use of such a stored platform profile **114** according to some embodiments will be described in detail below.

(43) FIG. 7 illustrates an embodiment in which the above-described posting platforms are online job boards and the posts are job requisitions. As described with respect to application platform **110**, application platform **710** may comprise an on-premise, cloud-based, or hybrid computing platform providing an execution platform and services to software applications. Job boards **720** may each comprise an on-premise, cloud-based or hybrid computing system which is integrated with recruiting posting application **730**. Job boards **720** may be selected to match various recruitment strategies of manager **740** of a given customer. In some embodiments, a single company or entity owns, controls or manages multiple job boards (e.g., accepting fees for allowing recruiters to place job openings on one of the job boards), and some embodiments include multiple companies or entities which own, control or manage a plurality of job boards that interact with application platform **710**.

(44) Generally, manager **740** may generate a job requisition using recruiting management application **750**. Via recruiting management application **750**, manager **740** may select one or more of job boards **720** on which to post the job requisition. Each of the selected job boards **720** may be associated with a respective board profile of board profiles **738** stored in data store **735**. At least one of the respective board profiles may have been generated and stored in data store **735** by a job board **720** in conjunction with integration service **732** as described with respect to process **200** and integration service **113**. Data store **735**, board profiles **738** and posts **736** may be examples of the more generic depictions of data store **115**, platform profiles **114** and posts **117** of FIG. 1.

(45) Each of board profile **738** of the selected job boards **720** may define fields whose attributes

indicate that the fields are required or optional, as will be described further in conjunction with the description of FIG. 8. Posting application 734 prompts manager 740 to provide values for these fields prior to transmitting the job requisition to the selected job boards. Then, for each selected job board of job boards 720, posting application 734 of recruiting posting component 730 formats and transmits the job requisition to the job board based on the board profile associated with the job board.

(46) Job boards 720 may communicate posted job requisitions to candidate 760 in any manner that is or becomes known. For example, candidate 760 may establish a subscription with a job board 720 to receive notices (e.g., via e-mail or text) of posted job requisitions which are associated with certain characteristics (e.g., particular location, particular position, particular degree requirement). Candidate 760 may also directly access a job board 720, with or without login credentials, to browse job requisitions posted thereto.

(47) According to some embodiments, candidate 760 may respond to a job requisition via a job board 720 to which the job requisition is posted. For example, the job board 720 may include a "Send resume and cover letter" hyperlink which candidate 760 may select to upload documents to recruiting marketing application 770 of platform 710. Recruiting marketing application 770 may be operated by manager 740 or other human resources personnel to facilitate the organization and review of communications received from candidates.

(48) Recruiting management application 750, posting application 734 and recruiting marketing application 770 may comprise elements of a suite of applications provided by platform 710. Manager 740 may access a home Web page of platform 710, provide suitable credentials, and be presented with an interface allowing selection of one of the applications. Each application may communicate with other applications of the suite via respective APIs.

(49) FIG. 8 is a flow diagram of process 800 to transmit a job posting to a job board according to some embodiments. The job board may be integrated with a job posting application as described above with respect to process 200. Process 800 may be implemented by processor-executable program code of a job posting application.

(50) A selection of a job requisition is received at S810. A job requisition may describe a job position and may include, for example, a title, a list of qualifications, a location, a salary, a list of benefits, and contact information. According to some embodiments, S810 comprises receipt of a selection of a job requisition from a list of previously-created job requisitions. In other embodiments, a job requisition may be generated from scratch without relying on a previously-created job requisition. In such embodiments, the selection of a job requisition occurs after performing text editing to create the job requisition.

(51) FIG. 9 is a view of UI 900 according to some embodiments. UI 900 may be displayed on a computing device operated by manager 740 of FIG. 7. In particular, manager 740 may operate the computing device to access a home page associated with recruiting management application 750 executing on platform 700. Manager 740 may navigate from the home page to a list of previously-stored job requisitions as shown in UI 900.

(52) Each of the job requisitions listed on UI 900 is associated with a checkbox 910 to allow selection thereof. Controls 920, 930 and 940 initiate a corresponding action upon a job requisition for which checkbox 910 is selected. According to some embodiments, selection of Post to Job Board control 940 results in reception at S810 of a selection of a job requisition to post to a job board. Embodiments are not limited to UI 900.

(53) Next, an instruction to transmit the selected job requisition to one or more job boards is received at S820. Continuing the above example, selection of control 940 of UI 900 may result in display of UI 1000 of FIG. 10. UI 1000 presents job title 1010 associated with the job requisition selected at S810. UI 1000 also presents several selectable job boards. According to some embodiments, the displayed job boards are each associated with a board profile 738 of data store 735.



(54) For example, selection of control **940** of UI **900** may cause recruiting management application **750** to transmit a request to posting application **734** (e.g., by calling a corresponding API). In response, posting application **734** identifies job boards from board profiles **738** and provides UI **1000** including the identified job boards. As described above, a board profile may include a logo or other descriptive information associated with a corresponding job board, and UI **1000** may present such a logo or information along with each presented job board.

(55) Manager **740** has selected three job boards in the FIG. **10** example. Manager **740** may then select Initiate Posting control **1020** to submit an instruction to posting application **734** to transmit the job requisition to the three selected job boards.

(56) Field definitions of each of the selected job boards are determined at **S830**. A field definition may include a name of a field and attributes associated with the field as described above. The field definitions of each selected job board may be stored in board profiles **738** and retrieved therefrom by posting application **734** in some embodiments. Moreover, the field definitions for at least one of the selected job boards may have been generated and stored by the job board as described with respect to process **200**.

(57) Based on the determined field definitions, a template is generated and presented at **S840**. The generated template includes fields for which values may or must be provided prior to posting the job requisition to the selected job boards. For example, each of the selected job boards may include a field Location which is associated with an attribute indicating that at least one value for the field must be provided. The template generated at **S840** therefore includes an input region for indicating a value for the field Location.

(58) As described above, a field may be associated with a pick-list of values. Two job boards may therefore be associated with a same field but the field may be associated with different pick-lists for each of the two job boards. In such a case, the template may be generated at **S840** to include an input region for the field which allows any value of either pick-list to be selected.

(59) In another example, one selected job board is associated with a field Salary and the other selected job boards are not associated with a field Salary. The template generated at **S840** may therefore include an input region for indicating a value for the field Salary, to ensure that this information is collected for the one selected job board.

(60) In some cases, one or more of the non-selected job boards may be associated with a given field while none of the selected job boards are associated with the given field. Accordingly, at **S840**, the generated template does not include an input region corresponding to the given field.

(61) Values input to the template are received at **S850**. FIG. **11** illustrates UI **1100** which may be generated and presented at **S840** according to some embodiments. UI **1100** includes input regions corresponding to fields Categories, Sub-Categories, Specialization, Track and Locations. Each of the fields is associated with a list of selectable values, and **S850** may comprise selection of certain ones of the selectable values. In some embodiments, one or more of the fields may include input controls allowing manual entry of one or more values for one or more given fields.

(62) **S850** may include evaluation of whether values have been input for all required fields. As mentioned above, a field may be required by one job board but optional for another job board. If both of such job boards are selected, the field may be considered required for purposes of the evaluation at **S850**, since it is required by at least one job board. If a user indicates completion of the template but values have not been input for all required fields, the user may be prompted to provide the missing values prior to proceeding to **S860**.

(63) At **S860**, a respective transfer protocol is determined for each of the selected job boards. The transfer protocol for a job board may be specified in the board profile associated with the job board, as described above. Next, at **S870**, the selected job requisition and the values input at **S850** are transmitted to each of the selected job boards, using the respective transfer protocol determined for each job board.

(64) The values transmitted to a particular job board may include less than all the values received at

**S850**, for example if some of the received values are associated with fields which are not associated with the particular job board, but with other selected job boards. According to some embodiments of **S870**, received values associated with a particular selected job board and the job requisition are formatted into a data structure (e.g., an XML feed) defined by a corresponding board profile created per process **200**, and the data structure is transmitted to the job board using a data transfer protocol which is also defined in the corresponding board profile. This process is repeated, successively or substantially in parallel, for each selected job board.

(65) FIG. **12** is a block diagram of a hardware system providing self-service platform integration according to some embodiments. Hardware system **1200** may comprise a computer server and may execute program code to perform any of the functions described herein. Hardware system **1200** may be implemented by a distributed cloud-based server, a virtual machine, or other implementation and may comprise an implementation of platform **110** in some embodiments. Hardware system **1200** may include other unshown elements according to some embodiments.

(66) Hardware system **1200** includes processing unit(s) **1210** operatively coupled to I/O device **1220**, data storage device **1230**, one or more input devices **1240**, one or more output devices **1250** and memory **1260**. I/O device **1220** may facilitate communication with external devices, such as third-party posting platforms, an external network, the cloud, or a data storage device. Input device(s) **1240** may comprise, for example, a keyboard, a keypad, a mouse or other pointing device, a microphone, knob or a switch, an infra-red (IR) port, a docking station, and/or a touch screen. Input device(s) **1240** may be used, for example, to enter information into hardware system **1200**. Output device(s) **1250** may comprise, for example, a display (e.g., a display screen) a speaker, and/or a printer.

(67) Data storage device **1230** may comprise any appropriate persistent storage device, including combinations of magnetic storage devices (e.g., magnetic tape, hard disk drives and flash memory), optical storage devices, Read Only Memory (ROM) devices, and RAM devices, while memory **1260** may comprise a RAM device.

(68) Data storage device **1230** stores program code executed by processing unit(s) **1210** to cause server **1200** to implement any of the components and execute any one or more of the processes described herein. Embodiments are not limited to execution of these processes by a single computing device. Data storage device **1230** may also store data and other program code for providing additional functionality and/or which are necessary for operation of hardware system **1200**, such as device drivers, operating system files, etc.

(69) The foregoing diagrams represent logical architectures for describing processes according to some embodiments, and actual implementations may include more or different components arranged in other manners. Other topologies may be used in conjunction with other embodiments. Moreover, each component or device described herein may be implemented by any number of devices in communication via any number of other public and/or private networks. Two or more of such computing devices may be located remote from one another and may communicate with one another via any known manner of network(s) and/or a dedicated connection. Each component or device may comprise any number of hardware and/or software elements suitable to provide the functions described herein as well as any other functions. For example, any computing device used in an implementation some embodiments may include a processor to execute program code such that the computing device operates as described herein.

(70) Embodiments described herein are solely for the purpose of illustration. Those in the art will recognize other embodiments may be practiced with modifications and alterations to that described above.

## Claims

1. A system comprising: a memory storing processor-executable program code; and at least one processor to execute the processor-executable program code to cause the system to: generate a link to an integration service application, the integration service application presenting a user interface to a user, the user interface including a smart template wizard to process one or more user inputs to generate a markup language template associated with a first posting platform; receive multiple markup language files from a first posting platform, each of the multiple markup language files conforming to a particular input data format of the first posting platform, the first posting platform comprising a platform on which information is capable of being posted for viewing by one or more users of the first posting platform; process the multiple markup language documents to automatically determine a first plurality of field definitions based on characteristics of the multiple markup language files, each field definition including a field and attributes associated with the field; transmit the first plurality of field definitions to the first posting platform; receive one or more modifications to the plurality of field definitions from the first posting platform; determine a transfer protocol for the first posting platform; generate a second plurality of field definitions based on the one or more modifications to the plurality of field definitions; store a modified plurality of field definitions, the modified plurality of field definitions comprising the second plurality of field definitions and unmodified field definitions from the first plurality of field definitions; receive an initial post from a user and a user selection indicating that the initial post is to be posted to the first posting platform; generate a formatted post, based on the initial post, the modified plurality of field definitions, and on the markup language template; and transmit the formatted post to the first posting platform via the identified transfer protocol.
2. The system according to claim 1, wherein the at least one processor is further to execute the processor-executable program code to cause the system to: receive a data transfer protocol from the first posting platform, wherein storage of the second plurality of field definitions comprises storage of the data transfer protocol in association with the second plurality of field definitions.
3. The system according to claim 1, wherein the at least one processor is further to execute the processor-executable program code to cause the system to: receive a second set of data from a second posting platform; determine a third plurality of field definitions based on the second set of data, each of the third plurality of field definitions including a field and attributes associated with the field; transmit the third plurality of field definitions to the second posting platform; receive a second one or more modifications to the third plurality of field definitions from the second posting platform; generate a fourth plurality of field definitions based on the second one or more modifications to the third plurality of field definitions; and store the fourth plurality of field definitions.
4. The system according to claim 3, wherein the at least one processor is further to execute the processor-executable program code to cause the system to: receive a first data transfer protocol from the first posting platform; and receive a second data transfer protocol from the second posting platform, wherein storage of the second plurality of field definitions comprises storage of the first data transfer protocol in association with the second plurality of field definitions, and wherein storage of the fourth plurality of field definitions comprises storage of the second data transfer protocol in association with the fourth plurality of field definitions.
5. The system according to claim 4, wherein the first set of data comprises a first extended Markup Language file, and wherein the second set of data comprises a second eXtended Markup Language file.
6. The system according to claim 1, wherein the first set of data comprises an extended Markup Language file.
7. The system according to claim 1, wherein the at least one processor is to further execute the processor-executable code to cause the system to: process a user input comprising a post to transmit to the first posting platform; format the post based at least in part on the second plurality

of field definitions; and transmit the formatted post to the first posting platform.

8. A method comprising: generating a link to an integration service application, the integration service application presenting a user interface to a user, the user interface including a smart template wizard to process one or more user inputs to generate a markup language template associated with a first posting platform; receiving multiple markup language files from a first posting platform, each of the multiple markup language files conforming to a particular input data format of the first posting platform, the first posting platform comprising a platform on which information is capable of being posted for viewing by one or more users of the first posting platform; processing the multiple markup language documents to automatically determine a first plurality of field definitions based on characteristics of the multiple markup language files, each field definition including a field and attributes associated with the field; transmitting the first plurality of field definitions to the first posting platform; receiving one or more modifications to the plurality of field definitions from the first posting platform; determining a transfer protocol for the first posting platform; generating a second plurality of field definitions based on the one or more modifications to the plurality of field definitions; storing a modified plurality of field definitions, the modified plurality of field definitions comprising the second plurality of field definitions and unmodified field definitions from the first plurality of field definitions; receiving an initial post from a user and a user selection indicating that the initial post is to be posted to the first posting platform; generating a formatted post, based on the initial post, the modified plurality of field definitions, and on the markup language template; and transmitting the formatted post to the first posting platform via the identified transfer protocol.

9. The method according to claim 8, further comprising: receive a data transfer protocol from the first posting platform, wherein storage of the second plurality of field definitions comprises storage of the data transfer protocol in association with the second plurality of field definitions, and wherein the first set of data transmitted to the first posting platform using the data transfer protocol.

10. The method according to claim 8, further comprising: receive a second set of data from a second posting platform; determine a third plurality of field definitions based on the second set of data, each of the third plurality of field definitions including a field and attributes associated with the field; transmit the third plurality of field definitions to the second posting platform; receive a second one or more modifications to the third plurality of field definitions from the second posting platform; generate a fourth plurality of field definitions based on the second one or more modifications to the third plurality of field definitions; and store the fourth plurality of field definitions.

11. The method according to claim 10, further comprising: receive a first data transfer protocol from the first posting platform; and receive a second data transfer protocol from the second posting platform, wherein storage of the second plurality of field definitions comprises storage of the first data transfer protocol in association with the second plurality of field definitions, and wherein storage of the fourth plurality of field definitions comprises storage of the second data transfer protocol in association with the fourth plurality of field definitions.

12. The method according to claim 8, wherein the first set of data comprises an extended Markup Language file.

13. The method according to claim 12, wherein the first set of data comprises a first eXtended Markup Language file, and wherein the second set of data comprises a second eXtended Markup Language file.

14. The method according to claim 8, further comprising: processing a user input comprising a post to transmit to the first posting platform; formatting the post based at least in part on the second plurality of field definitions; and transmitting the formatted post to the first posting platform.

15. A system comprising: a memory storing processor-executable program code; a data store; and at least one processor to execute the processor-executable program code to cause the system to: generate a link to an integration service application, the integration service application presenting a

user interface to a user, the user interface including a smart template wizard to process one or more user inputs to generate a markup language template associated with a first posting platform; receive a first post and an instruction, from a user, to transmit the first post to a first posting platform, the first posting platform comprising a platform on which information is capable of being posted for viewing by one or more users of the first posting platform; receive multiple markup language files from a first posting platform, each of the multiple markup language files conforming to a particular input data format of the first posting platform, the first posting platform comprising a platform on which information is capable of being posted for viewing by one or more users of the first posting platform; process the multiple markup language documents to automatically determine a first plurality of field definitions associated with the first posting platform and stored in the data store based on characteristics of the multiple markup language files, each of the first plurality of field definitions including a field and attributes associated with the field; determine a transfer protocol for the first posting platform; format the first post according to the first plurality of field definitions and on the markup language template; transmit the formatted first post to the first posting platform via the identified transfer protocol; receive a second post and a second instruction, from the user, to transmit the second post to a second posting platform; determine a second plurality of field definitions associated with the second posting platform and stored in the data store, each of the second plurality of field definitions including a field and attributes associated with the field; format the second post according to the second plurality of field definitions; and transmit the formatted second post to the second posting platform.

16. The system according to claim 15, the at least one processor to execute the processor-executable program code to cause the system to: determine a first data transfer protocol associated with the first posting platform and stored in the data store; and determine a second data transfer protocol associated with the second posting platform and stored in the data store, wherein the formatted first post is transmitted to the first posting platform according to the first data transfer protocol, and wherein the formatted second post is transmitted to the second posting platform according to the second data transfer protocol.

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